

Astro HW 12

Pierson Lipschultz

December 3, 2025

12.1 Eddington luminosity and massive stars

a)

Given ...

$$L = 1.2L_{\odot} \left(\frac{M}{M_{\odot}} \right)^{3.92}$$
$$L_{edd} = \frac{4\pi GM_p c}{\sigma_e}$$
$$\sigma_e = 6.55 \times 10^{-29} m^2, \quad m_p = 1.67 \times 10^{-27} kg$$

$$M_{max} = 3.514 \times 10^1 L_{\odot}$$

b)

This calculated value of $M_{max} = 3.514 \times 10^1 L_{\odot}$ is much lower than a considerable amount of observed systems. For example, the current most massive star has a proposed mass of $346 \pm 42 M_{\odot}$ [1]. All the most massive stars are Wolf-Rayet stars, meaning that they are so massive that they start to shed their exterior hydrogen envelopes.

12.2 Accretion disk temperatures

a)

Given ...

$$T \approx \left[\frac{GM}{2\sigma_{SB} r^2} \Sigma v \right]^{1/4}$$
$$\dot{M} = 2\pi r \Sigma(r) v(r)$$

We find ...

$$\dot{M} = 2\pi r \Sigma(r) v(r)$$
$$\Sigma(r) v = \frac{\dot{M}}{r 2\pi}$$
$$T \approx \left[\frac{GM}{2\sigma_{SB} r^2} \frac{\dot{M}}{r 2\pi} \right]^{1/4}$$
$$T \approx \left[\frac{GM \dot{M}}{4\pi \sigma_{SB} r^2} \right]^{1/4}$$

b)

Given ...

$$L_e = 4\pi r^2 \sigma_{sb} T^4$$

$$r = \frac{6GM}{c^2}$$

$$r \propto M, \quad M \propto L_E$$

We can find ...

$$L_e = 4\pi \left(\frac{6GM}{c^2} \right)^2 \sigma_{sb} T^4$$

By proportionality ...

$$\Rightarrow M^2 T^4 \propto M$$

$$\Rightarrow T^4 \propto \frac{M}{M^2}$$

$$\boxed{T = M^{-1/4}}$$

c)

Given ...

$$\dot{M} = \dot{m} \left(\frac{4\pi G M m_p}{\sigma_e \eta c} \right)$$

$$\dot{m} = 1 \quad \eta \sim .1$$

$$r = \frac{(6GM)}{c^2}$$

$$T(r) \approx \left[\frac{G M \dot{M}}{4\pi \sigma_{SB} r^3} \right]^{1/4}$$

We find ...

$$\boxed{T_{4M_\odot} = 1.755 \times 10^7 \text{ K}, \quad T_{10^9 M_\odot} = 1.396 \times 10^5 \text{ K}}$$

d)

We find that they lie in the microwave and radio wave spectrum.

12.3 Stellar close encounters

a)

Given ...

$$R_{eff} = \frac{6G}{v^2}$$

$$M_{star} = M_\odot, \quad v = 30 \text{ km/s}$$

We find ...

$$\boxed{R_{eff} = 8.847 \times 10^{11} \text{ m}}$$

b)

Given ...

$$t = \frac{1}{n_* v (\pi R_{eff}^2)}$$

$$M_{star} = M_{\odot}, \quad v = 30 \text{ km/s}, \quad n_* = .1^{-3} \text{ pc}$$

We find ...

$$t_{col} = 1.262 \times 10^{14} \text{ yr}$$

c)

$$P_{col} = 5 \times 10^9 \text{ yr} / t_{col} \approx 3.962 \times 10^{-5}$$

12.4 Expansion of a Newtonian universe

a)

Given ...

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G \rho^{c,0}}{3} \frac{1}{a^3}$$

We can calculate ...

$$\dot{a}^2 = \frac{8\pi G \rho^{c,0}}{3} \frac{1}{a}$$

$$\frac{8\pi G \rho^{c,0}}{3} = k$$

$$\dot{a} = \sqrt{\frac{k}{a}}$$

$$\frac{da}{dt} = \sqrt{\frac{k}{a}}$$

$$da = \sqrt{\frac{k}{a}} dt$$

$$\sqrt{a} da = \sqrt{k} dt$$

$$\int \sqrt{a} da = \int \sqrt{k} dt$$

$$\frac{2}{3} a^{3/2} = t \sqrt{k}$$

$$a = \left(\frac{3}{2} t \sqrt{k} \right)^{2/3}$$

$$\boxed{a(t) = \left(\frac{3}{2} t \sqrt{\frac{8\pi G \rho_{c,0}}{3}} \right)^{2/3}}$$

b)

c)

Given ...

$$\frac{1}{t_0} = \frac{1}{H_0}$$

$$H_0 = 70 \text{ km/s/Mpc}$$

We find ...

$T_{uni} = 13.97 \text{ Gyr}$

12.5 Olbers's paradox again

a)

Given ...

$$n_{wd} = \frac{3H_0^2}{8\pi G}$$

We find

$n_{wd} = 6.613 \times 10^{-57} \text{ m}^3$
--

b)

Given ...

$$n_{wd} = 6.613 \times 10^{-57} \text{ m}^3$$

$$\sigma = \pi r_{WD}^2$$

$$\ell = \frac{1}{n_{wd}\sigma}$$

We find...

$$\ell = 9.946 \times 10^{41} \text{ m}$$

c)

Because the distance till one sees a WD is much greater than the size of the observable universe, it does place an upper limit on the number density.

12.6

Yes! If we do it with basketballs we find ...

$$n_{basketball} = 1.534 \times 10^{-26} \text{ 1/m}^3 \quad \ell_{basketball} = 1.441 \times 10^{27} \text{ m}$$

Sources

- [1] Z. Keszthelyi et al. "Evolutionary models for the very massive stars in the R136 cluster of 30 Doradus in the Large Magellanic Cloud". In: *Astronomy and Astrophysics* 700 (Aug. 2025), A186. ISSN: 1432-0746. DOI: 10.1051/0004-6361/202554758. URL: <http://dx.doi.org/10.1051/0004-6361/202554758>.